Clayton Ryan Ulm

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Education

Ph.D. in Health Behavior 08/2025 – Current
Advisor: Dr. Sarah Mills
Gillings School of Global Public Health, University of North Carolina at Chapel Hill

Pre-doctoral Trainee, Biosocial Training Program 08/2026 – Current
Mentors: Drs. Sarah Mills and Kenneth Bollen
Carolina Population Center, University of North Carolina at Chapel Hill

MSPH in Health Behavior 08/2023 – 12/2025
Advisor: Dr. Sarah Mills
"State-Level Trends in Urban-Rural Differences in Cigarette Smoking in the United States"
Gillings School of Global Public Health, University of North Carolina at Chapel Hill

B.S. in Psychology, Summa Cum Laude, GPA: 3.93 08/2018 – 05/2022
"Statistical numeracy and belief bias: A case of medical decision making"
Advisor: Dr. Edward Cokely
University of Oklahoma

Research Experience

Data Analyst 11/2025 – Current
Stanford University
Stanford Food Policy Lab, Department of Health Policy, School of Medicine
- Mixed-effect analyses and data visualization for research on food policy and social media warning labels.

Graduate Research Assistant 08/2023 – Current
University of North Carolina at Chapel Hill
Department of Health Behavior, Gillings School of Global Public Health
- Causal loop diagrams, meta-analyses, and regression analyses on tobacco policy and health behaviors.

Staff Research Assistant 01/2022 – 05/2023
University of Oklahoma Health Sciences Center
Health Promotion Research Center, Stephenson Cancer Center
- Mixed-effect analyses of predictors of app engagement; participant recruitment and qualitative interviews.

Research Assistant x3 (for 3 separate labs) 01/2020 – 05/2022
University of Oklahoma
(1) Decision Analytics Lab; (2) Ethical Choice Education Lab; (3) Day Lab, Department of Psychology
- Structural equation modeling on statistical numeracy affecting medical decision-making.

Peer-reviewed Publications

6. Grummon, A. H., O'Sullivan, K., Zeitlin, A. B., Taillie, L. S., Lee, C. J. Y., Walker, A. J., Tiefenbeck, V., Fleisch, E., **Ulm, C.**, & Robinson, T. N. (2026). Effect of food swap recommendations on the healthfulness and carbon footprint of grocery purchases in a naturalistic online store: A randomized parallel trial. *PLOS Medicine*, *23*(8), e1004847. https://doi.org/10.1371/journal.pmed.1004847

5. Grummon, A. H., **Ulm, C.**, Zeitlin, A. B., Hall, M. G., Falbe, J., Kline, N. A., & Robinson, T. N. (2026). Designing social media health warnings for teenagers and young adults: A randomized clinical trial. *JAMA Health Forum,* 7(9), e263014. https://doi.org/10.1001/jamahealthforum.2026.3014

4. Langford, J. S., Hébert, E., Jones, D., Tonkin, S., Barker, B., **Ulm, C.**, Shi, D., Becerra, J., Businelle, M. (2026). Effects of Ecological Momentary Assessment Prompting Schedule on Affect Measurement Variability and Associations with Next-Day Health Behaviors. *Assessment.* Advance online publication.https://doi.org/10.1177/10731911261444953

3. **Ulm, C.**, Hill, F., Mills, S. D., Chelius, C., & Hecht, E. M. (2026). Association between unhealthy food consumption and mental health outcomes in children and adolescents: an umbrella review. *Child and Adolescent Psychiatry and Mental Health*, *20*(1), 70. https://doi.org/10.1186/s13034-026-01079-4

2. Hill, F., Hassmiller Lich, K., Wilson, M., **Ulm, C.**, Golden, S. D., Ribisl, K. M., Baldwin-SoRelle, C., & Mills, S. D. (2026). The public health impacts of flavored cigar bans: A systematic review and meta-analysis. *Nicotine & Tobacco Research*, *28*(6), 889–900. https://doi.org/10.1093/ntr/ntag014

1. **Ulm, C.**, Chen, S., Fleshman, B., Benson, L., Kendzor, D. E., Frank-Pearce, S., Neil, J. M., Vidrine, D., & Businelle, M. S. (2024). Smartphone-based survey and message compliance in adults initially unready to quit smoking: secondary analysis of a randomized controlled trial. *JMIR Formative Research*, *8*(1), e56003. https://formative.jmir.org/2024/1/e56003/

Under Review

10. **Ulm, C.**, Mills, S. D., Wilson, M., Hill, F., Liebgott, E., Donnelly, A., Golden, S. D., Ribisl, K. M., & Lich, K. H. [Submitted for peer review]. Understanding causes and effects of menthol cigarette bans: a systematic evidence synthesis using causal loop diagramming. *PLOS One.*

9. Stanfield Gamble, A., Mayorga, M. E., O'Leary, M. C., Puppala, S. S., **Ulm, C.**, & Lich, K. H. [Revise and resubmit]. A Systematic Review of Metamodeling in Health and Recommendations for Cancer Research and Practice. *Preventive Medicine.*

8. **Ulm, C.**, Chelius, C., Zeitlin, A. B., & Grummon, A. H. [Submitted for peer review]. Psychological factors associated with support for SNAP restrictions. *Preventive Medicine.*

7. Huang, Y., **Ulm, C.**, Zeitlin, A. B., Falbe, J., Robinson, T. N., & Grummon, A. H. [Revise and resubmit]. Effects of ultra-processed food warnings on social media posts among teens and young adults. *JAMA Network Open.*

6. Zeitlin, A. B., Chelius, C., Chow, G., Robinson, T. N., Petimar, J., Halpern-Felsher, B., Roberto, C. A., **Ulm, C.**, & Grummon, A. H. [Submitted for peer review]. Exploring young adults' reactions to food marketing tactics to inform counter-marketing campaigns: A qualitative study. *Journal of the Academy of Nutrition and Dietetics.*

5. Montgomery, A., Tonkin, S. S., Gajos, J. M., Gonzalez, J., Barrett, Z. C. W., De La Torre, I., **Ulm, C.**, Cannell, M. B., Robison, J., Kendzor, D. E., & Businelle, M. S. [Submitted for peer review]. Psychotropic medication non-adherence and its impact on mental health and emergency room visits among justice system involved adults experiencing homelessness. *Journal of Health Care for the Poor and Underserved.*

4. Grummon, A. H., **Ulm, C.**, Kline, N. A., Hall, M. G., Zeitlin, A. B., & Robinson, T. N. [Submitted for peer review]. Using large language models to develop and test social media warnings. *JAMA Network Open.*

3. **Ulm, C.**, Mills, S. D., Wilson, M., Hill, F., Liebgott, E., Donnelly, A., Golden, S. D., Ribisl, K. M., & Lich, K. H. [Submitted for peer review]. Improving the effectiveness of flavored cigar sales restrictions: a systems science approach. *Nicotine & Tobacco Research.*

2. **Ulm, C.**, Golden, S. D., Hill, F., Wiesen, C. A., & Mills, S. D. [Revise and resubmit]. State-Level Trends in Urban-Rural Differences in Cigarette Smoking in the United States. *PLOS One*. Preprint posted to medRxiv (medRxiv 2026.08.06.26359850) 8/10/26.

1. O'Leary, M. C., Koutouan, P. R., Stanfield, A., Mayorga, M. E., Sharma, K., **Ulm, C.**, DeGroff, A., Richardson, L. C., & Lich, K. H. [Submitted for peer review]. Using simulation to assess the potential impact and cost-effectiveness of increased follow-up colonoscopy completion after positive stool tests among medically underserved populations in public health screening initiatives.

In Preparation

3. **Ulm, C.**, Hassmiller Lich, K., & Mills, S. D. [In preparation]. Within-Person Associations of E-cigarette and Cigarette Smoking and Relative Harm Perceptions.

2. **Ulm, C.**, Mills, S. D. [In preparation]. Constructing an immune/inflammatory health score to predict follow-up cognitive performance among US midlife adults.

1. Zhang, L., Gu, Y., Mills, S. D., Lich, K. H., **Ulm, C.** [In preparation]. Systems Literature Analysis Engine: Automating the connection of scientific evidence to inform causal loop diagrams.

Conference Presentations

16. Bland, T., **Ulm, C.**, Xiao, H., Beitscher, C., Mills, S. D., & Hassmiller Lich, K. (2025, August). Systems Literature Analysis Engine: Automating the connection of scientific evidence to inform causal loop diagrams. Work-in-progress presentation at the 43rd International System Dynamics Conference, Boston, MA

15. **Ulm, C.**, Liebgott, E., Donnelly, A., Wilson, M., Hill, F., Mills, S. D., Golden, S. D., Ribisl, K. M., & Hassmiller Lich, K. (2025, August). Contextual factors influencing flavored tobacco ban effectiveness: An empirical evaluation of industry tactics and policy enforcement. Panel presentation at the National Conference on Tobacco or Health, Chicago, IL.

14. Mills, S. D., Hill, F., **Ulm, C.**, Wilson, M., Liebgott, E., Donnelly, A., Golden, S. D., Ribisl, K. M., & Hassmiller Lich, K. (2025, August). Improving the design and implementation of local and state policies that ban the sale of flavored tobacco products. Poster presentation at the National Conference on Tobacco or Health, Chicago, IL.

13. Hill, F., Hassmiller Lich, K., Wilson, M., **Ulm, C.,** Golden, S. D., Ribisl, K. M., & Mills, S. D. (2025, March). The public health impacts of flavored cigar bans: A systematic review and meta-analysis. Oral presentation at the annual meeting of the Society for Research on Nicotine and Tobacco, New Orleans, LA.

12. **Ulm, C.**, Golden, S. D., Hill, F., Wiesen, C. A., & Mills, S. D. (2025, March 14). State-level trends in urban-rural smoking disparities from 2018 to 2022. Poster presentation at the annual meeting of the Society for Research on Nicotine and Tobacco, New Orleans, LA.

11. Hassmiller Lich, K., **Ulm, C.**, Wilson, M., Hill, F., Golden, S. D., Ribisl, K. M., & Mills, S. D. (2024, October). Incorporating context into simulation models of tobacco policy. Oral presentation at the Tobacco Regulatory Science Grantee Meeting, Bethesda, MD.

10. Montgomery, A., Tonkin, S., Cannell, M. B., Gonzalez, J., De La Torre, I., **Ulm, C.**, Robison, J., Kendzor, D. E., & Businelle, M. S. (2024, March 13–16). Medication non-adherence and its impact on mental health and substance use among adults experiencing homelessness. Poster presentation at the Society of Behavioral Medicine, Philadelphia, PA.

9. **Ulm, C.**, Hecht, E. (2023, September 28). The interrelationships of unhealthy food consumption and morbidities in youth: An umbrella review. Poster presentation at the 2023 North Carolina Public Health Association Conference, Concord, NC.

8. Cho, J., **Ulm, C.**, & Cokely, E. (2023, July 29). Risk literacy is associated with reduced belief bias in medical risk evaluations: Evidence from structural modeling of cognitive abilities and process tracing. Poster presentation at the annual meeting of the Cognitive Science Society, Sydney, Australia.

7. **Ulm, C.**, Bain, C., Flores, I., Cheney, M., Zingg, T., Kendzor, D., Frank-Pearce, S., Kezbers, K., & Businelle, M. (2023, May 1). Barriers to survey completion in a cessation app for smokers unmotivated to quit: A qualitative study. Poster presentation at the 2023 Colloquium on Qualitative Health Sciences Research in Oklahoma, Oklahoma City, OK.

6. **Ulm, C.**, Chen, S., Fleshman, B., De La Torre, I., Benson, L., Hébert, E., Kendzor, D., Frank-Pearce, S., Neil, J., Song, H., & Businelle, M. (2023, March 10). Do smokers initially unmotivated to quit open smartphone messages that promote and support smoking cessation attempts? Poster presentation at the 2023 Annual Cancer Research Symposium, Oklahoma City, OK.

5. **Ulm, C.**, Gajos, J., Becerra, J., Barry, A., Hébert, E., Ra, C., Shi, D., & Businelle, M. (2023, April 26–29). Private religiosity better predicts less hazardous drinking than public religiosity. Poster presentation at the Society of Behavioral Medicine, Phoenix, AZ.

4. Becerra, J., Shi, D., Gajos, J., Barker, B., **Ulm, C.**, De La Torre, I., Hébert, E., Piper, M., & Businelle, M. (2023, April 26–29). Are there benefits to using fixed versus random or 2 versus 4 smartphone-based ecological momentary assessments? Poster presentation at the Society of Behavioral Medicine, Phoenix, AZ.

3. De La Torre, I., Shi, D., **Ulm, C.**, Walters, D., Hébert, E., & Businelle, M. (2023, April 26–29). Cannabis use is related to increased physical activity. Poster presentation at the Society of Behavioral Medicine, Phoenix, AZ.

2. **Ulm, C.**, Cho, J., & Cokely, E. (2022, April 3). Statistical numeracy and belief bias: A case of medical decision making. Poster presentation at the 2022 Psychology Graduate Student Association, Norman, OK.

1. **Ulm, C.**, Kezbers, K., Benson, L., Kendzor, D., Frank-Pearce, S., Neil, J., De La Torre, I., & Businelle, M. (2022, March 4). Relationship between sleep quality and readiness to change weight-related health behaviors in smokers unmotivated to quit. Poster presentation at the 2022 Annual Cancer Research Symposium, Oklahoma City, OK.

Professional Experiences, Services & Memberships

Scientific Reports	Peer Reviewer	08/2026 – 10/2026
Preventing Chronic Disease	Peer Reviewer	04/2026 – 04/2026
UNC Tobacco Center of Regulatory Science	Trainee	08/2023 – Current
International Systems Dynamics Conference	Peer Reviewer	04/2025 – 04/2025
Society for Research on Nicotine and Tobacco	Member	03/2025 – Current
North Carolina Public Health Association	Member	05/2023 – 05/2024
Association of the United States Army	Member	05/2023 – Current
Institute of Etiological Research	Fellow	02/2023 – 05/2025
Society of Behavioral Medicine	Member	12/2022 – 12/2023

Undergraduate Research Journal	Peer Reviewer	06/2021 – 08/2023
Research Society on Alcoholism	Mentee	12/2021 – 08/2023

Honors & Scholarships

Graduate Scholarships x4 ($20,000)	EOD Warrior Foundation	2023-2027
Leon Van Autreve Scholarship ($2,000)	Association of the United States Army	2025-2026
Patriot Foundation Scholarships x2 ($18,700)	NC Patriot Foundation	2024-2026
American Legion Legacy Scholarship x3 ($10,000)	American Legion	2023-2026
Department Travel Award ($684)	Health Behavior Department UNC-CH	2025
Folds of Honor Scholarship x4 ($13,750)	Folds of Honor	2023-2027
Gary and Carolyn Koch Merit Scholarship	University of North Carolina Chapel Hill	2023-2025
Public Health Student Scholarship	University of North Carolina Chapel Hill	2023-2025
Johansson Scholarship	University of North Carolina Chapel Hill	2023-2025
Inclusive Excellence Scholarship	University of North Carolina Chapel Hill	2023-2025
MilDef Way Scholarship ($5,000)	MilDef/Association of the United States Army	2024
AUSA National Scholarship ($5,000)	Association of the United States Army	2023
Institute Student Investigator Grant ($4,000)	Institute of Etiological Research	2023
President's Honor Roll	University of Oklahoma	2018-2022
Dean's Honor Roll	University of Oklahoma	2018-2022
Work Assistance Tuition Waiver x3 ($12,000)	University of Oklahoma	2018-2022
Distinguished Scholar ($10,000)	University of Oklahoma	2018-2022
DEA Program Scholarship ($46,728)	U.S. Department of Veterans Affairs	2018-2022
Rural Scholarship Awardee ($48,000)	Hagen Scholarship Foundation	2018-2022
Study Abroad Scholarship ($8,000)	Hagen Scholarship Foundation	2021
Value Investing Training Scholarship ($17,000)	Hagen Scholarship Foundation	2019
Arch and Jo Gilbert Endowed Scholarship ($1,250)	University of Oklahoma	2019
Undergraduate Scholarship ($2,000)	EOD Warrior Foundation	2017